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 **AKI 2026**

Better than classical ?

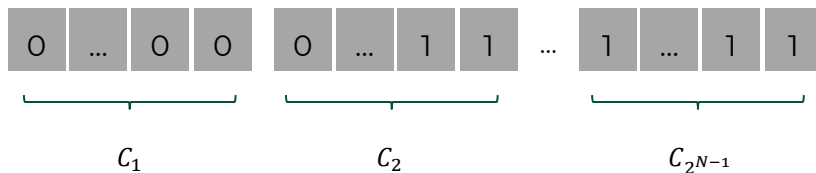
Quantum Temporal Convolution Network

Rihab Hoceini and Ahmed Bouida

Classical Computation Limits

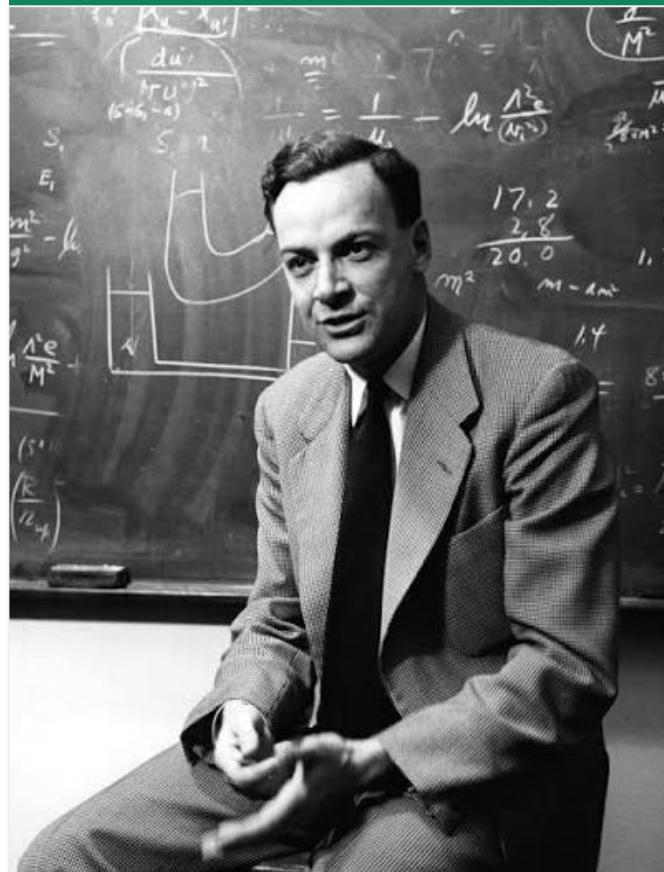
Nature **does not** behave like a binary switch

All possible configurations of System of N Components



Total number of
configuration

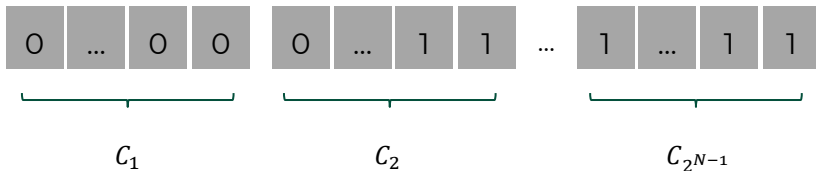
$$2^N$$



Classical Computation Limits

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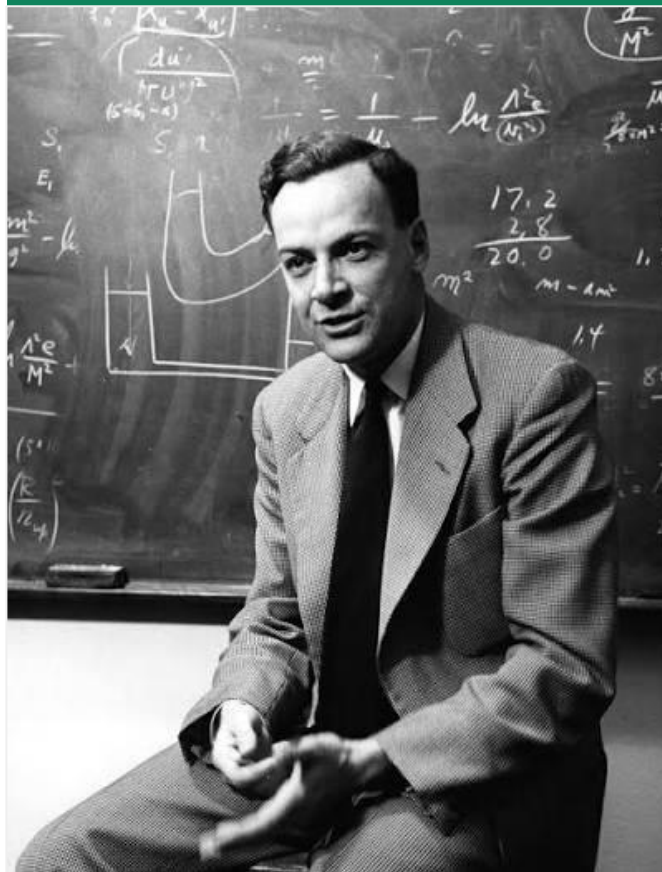
But with quantum

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} c_x |x\rangle$$

Total number of
configuration

2^N amplitudes

In one state

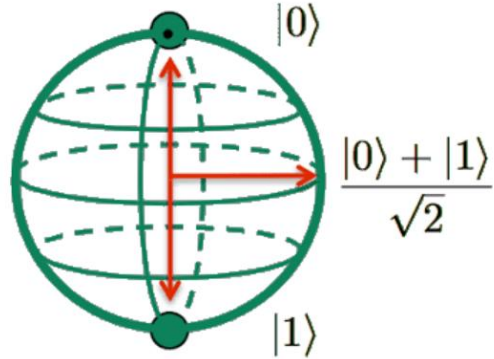


Quantum Properties in computation

● 0

● 1

Classical bit



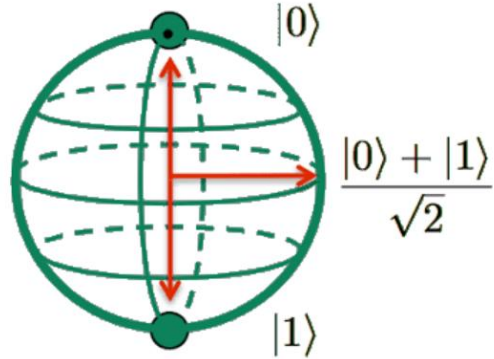
Quantum bit

Quantum Properties in computation

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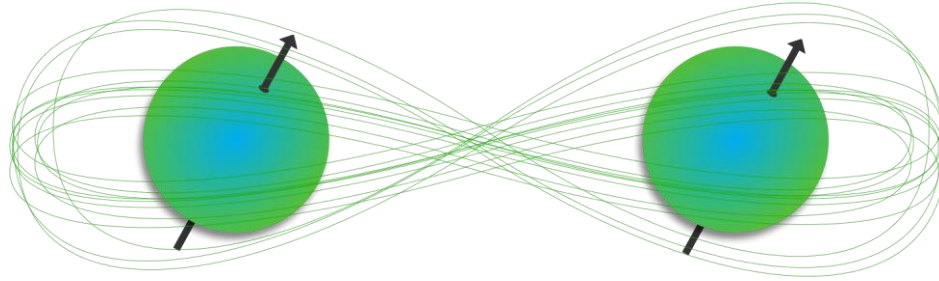
Classical bit



Quantum bit

Superposition

Quantum Properties in computation



Quantum entanglement creates a strong correlation between qubits in such systems. Measuring one qubit immediately determines the state of the other.

Efficient solutions

Efficient algorithms scale polynomially with input size

Shor's Algorithm

Can factor integer numbers in **polynomial time** on a quantum computer faster than the best known classical algorithm

Shor IEEE FOCS 1994

Grover's Algorithm

A quantum algorithm for searching an unsorted database with N entries **Quadratic speedup** compared to classical computer.

Grover ACM STOC 1996

The Problem

The Quantum Tortoise and the Classical Hare: A simple framework for understanding which problems quantum computing will accelerate (and which it will not)

[Sukwoong Choi](#), [William S. Moses](#), [Neil Thompson](#)

Quantum computing promises transformational gains for solving some problems, **but little to none for others**. For anyone hoping to use quantum computers now or in the future, it is important to know which problems will benefit. In this paper, we introduce a framework for answering this question

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Which problems should actually be solved
Better using quantum computing?



“Since the birth of quantum computing, researchers have been looking for the best place to apply quantum algorithms.”

Determining the Best Applications for Quantum Machine Learning (ICLR 2022)

Benchmarking quantum machine learning models

Better than classical? The subtle art of benchmarking quantum machine learning models

Joseph Bowles,^{1,*} Shahnawaz Ahmed,^{1,2,†} and Maria Schuld^{1,‡}

¹*Xanadu, Toronto, ON, M5G 2C8, Canada*

²*Chalmers University of Technology*

(Dated: March 15, 2024)

Benchmarking models via classical simulations is one of the main ways to judge ideas in quantum machine learning before noise-free hardware is available. However, the huge impact of the experimental design on the results, the small scales within reach today, as well as narratives influenced by the commercialisation of quantum technologies make it difficult to gain robust insights. To facil-

Fair benchmarking should encompass multiple dimensions including :
data encoding , model design , data complexity , optimization

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TCN Model
Temporal Convolution network

Research Gaps

Several studies introduce Quantum TCN architectures.
A fair comparison with classical TCNs is **still lacking**.

Paper	Proposed Quantum model	Compared against
Ahmad and Jas, 2025	QTCN	QCNN
Zhuang et al., 2025	Bert-QTCN	Bert-TCN
Yang et al., 2022	QTCN-LSTM	TCN-LSTM

Research Gaps

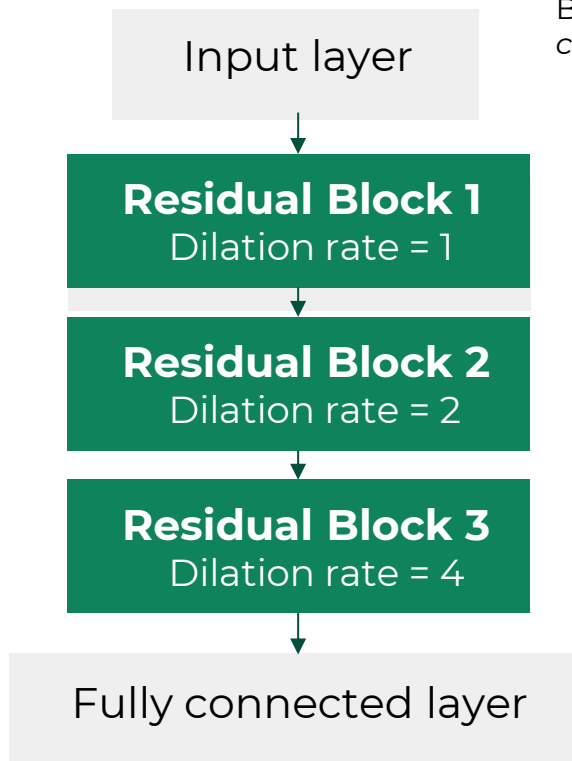
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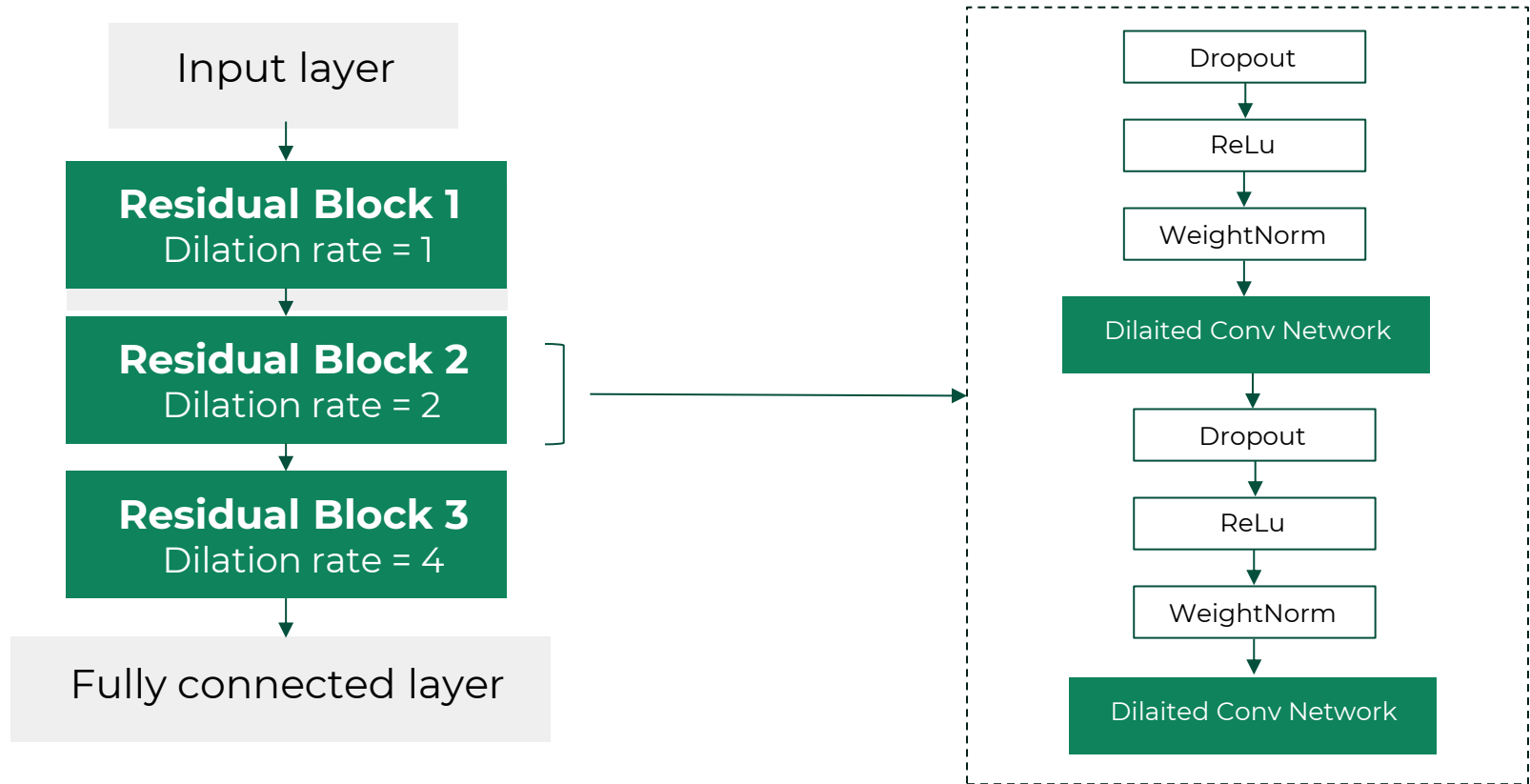
"Better than classical?" remains an open question

TCN structure

Bai et al. (2018). *An empirical evaluation of generic convolutional and recurrent networks for sequence modeling.*



TCN structure

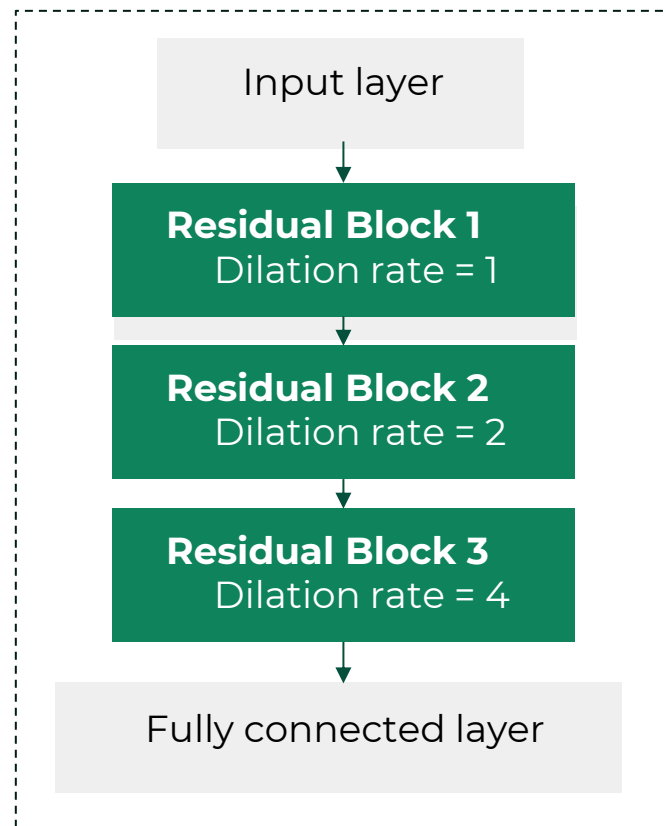


What could be the best strategy to integrate quantum

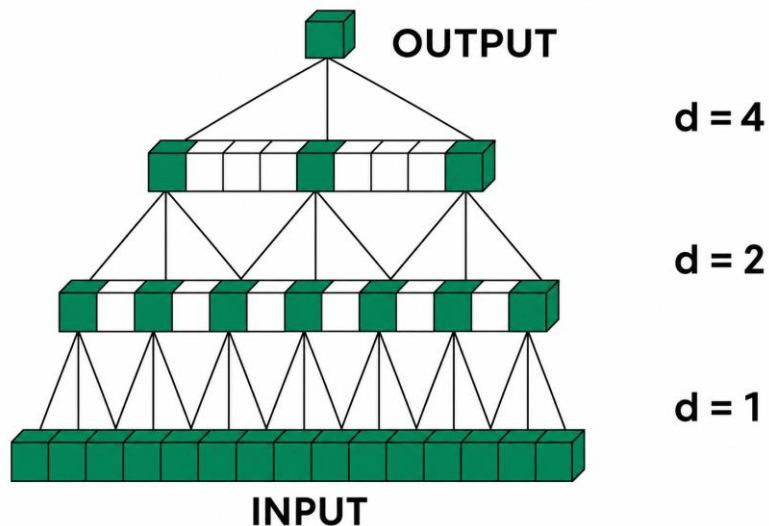
B. Hybrid Quantum-Classical Models

Near-term quantum processors are still fairly small and noisy, thus quantum models cannot disentangle and generalize quantum data using quantum processors alone. NISQ processors will need to work in concert with classical co-processors to become effective. We anticipate that investigations into various possible *hybrid quantum-classical machine learning algorithms* will be a productive area of research and that quantum computers will be most useful as hardware accelerators, working in symbiosis with traditional computers.

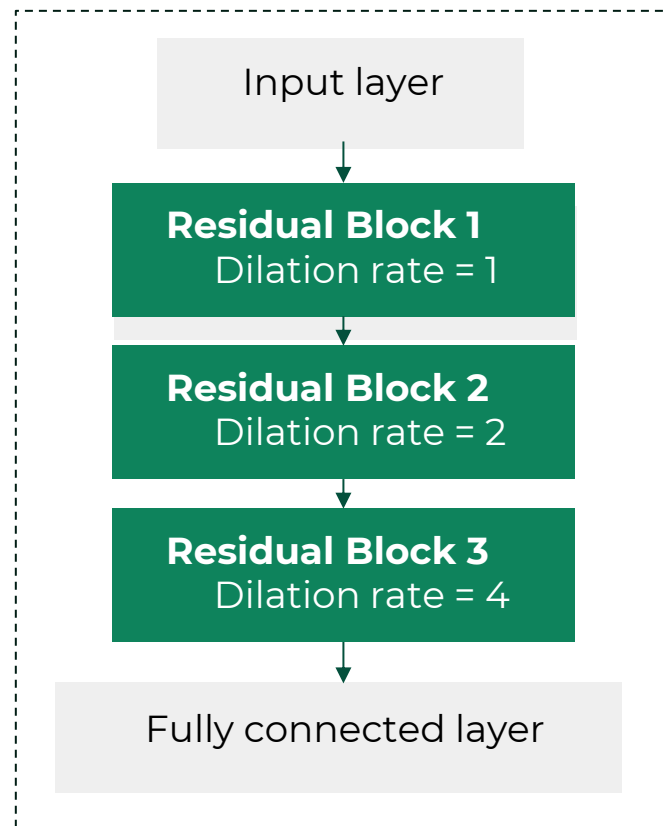
Broughton et al., TensorFlow Quantum, 2021



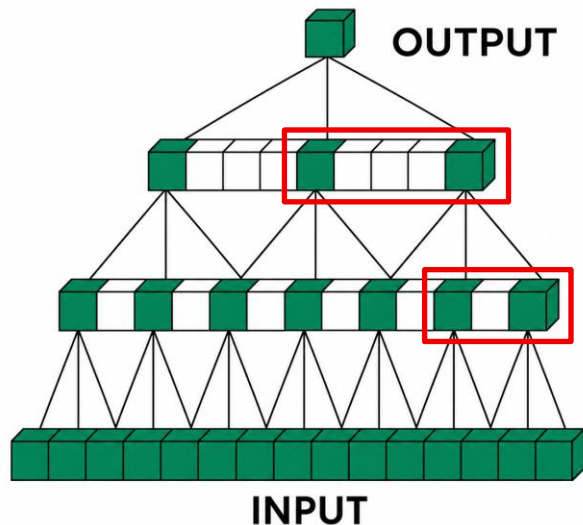
Dilated Convolution network



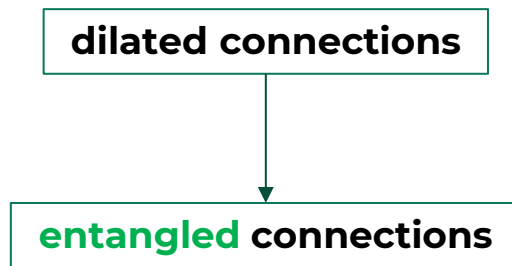
Yu & Koltun (2016). Multi-Scale Context Aggregation by Dilated Convolutions.



Dilated Convolution network



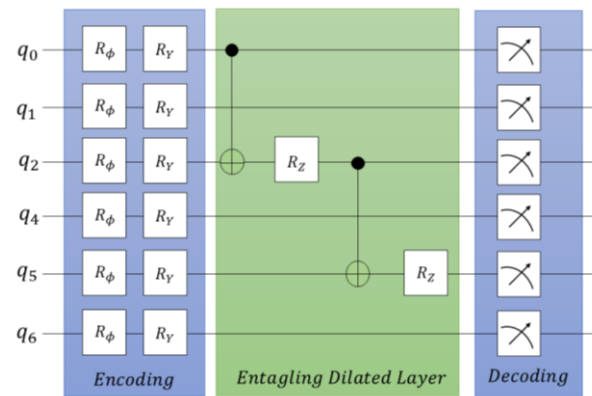
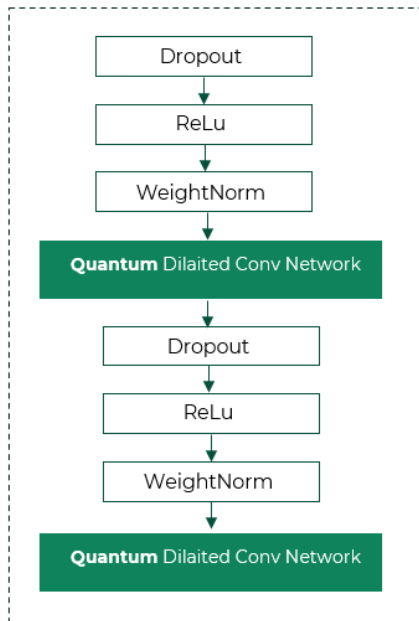
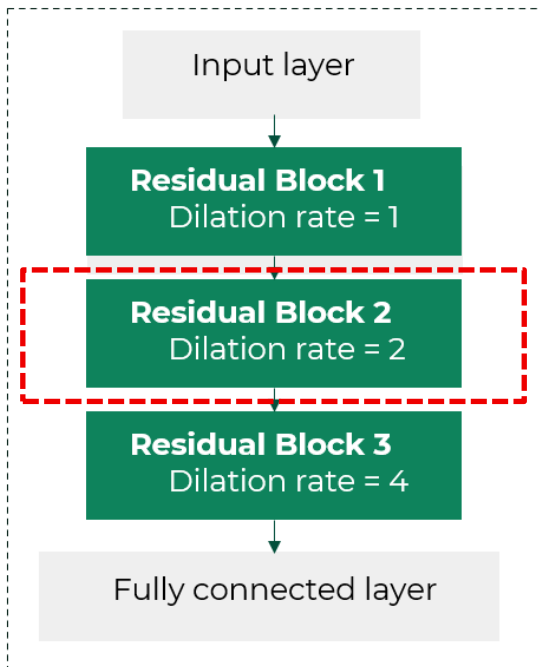
what does the **quantum** version look like



Entanglement between
Qubits i and $i + d$ for even i

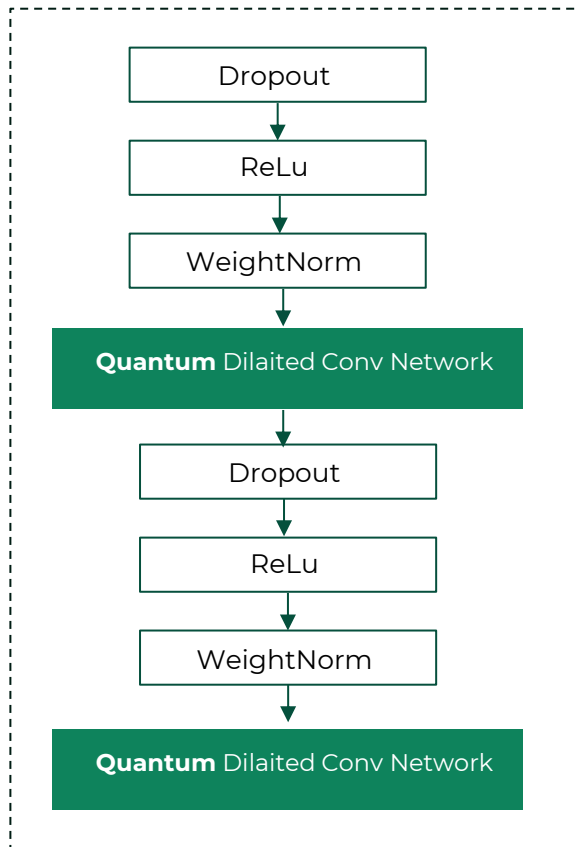
Yu & Koltun (2016). Multi-Scale Context Aggregation by Dilated Convolutions.

Quantum TCN

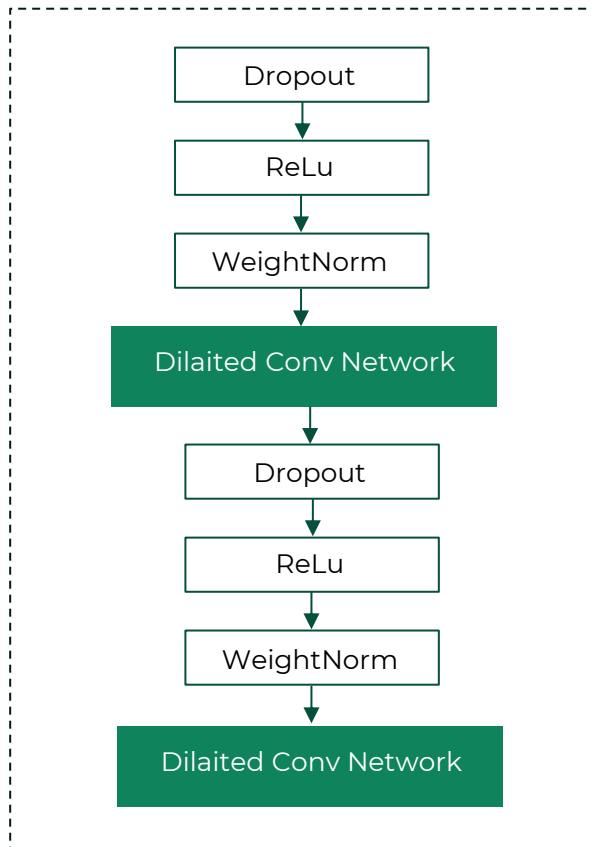


QDCNN circuit with
dilation rate=2

Quantum TCN

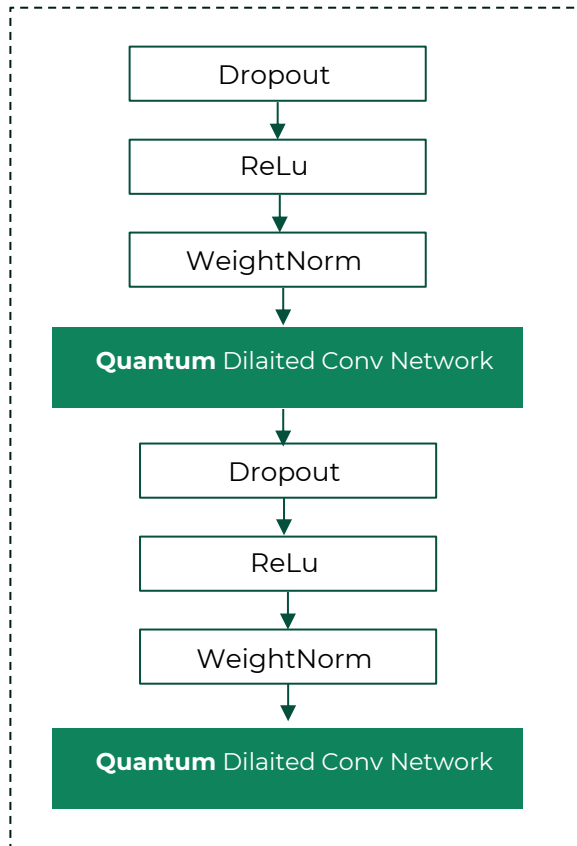


Classical TCN

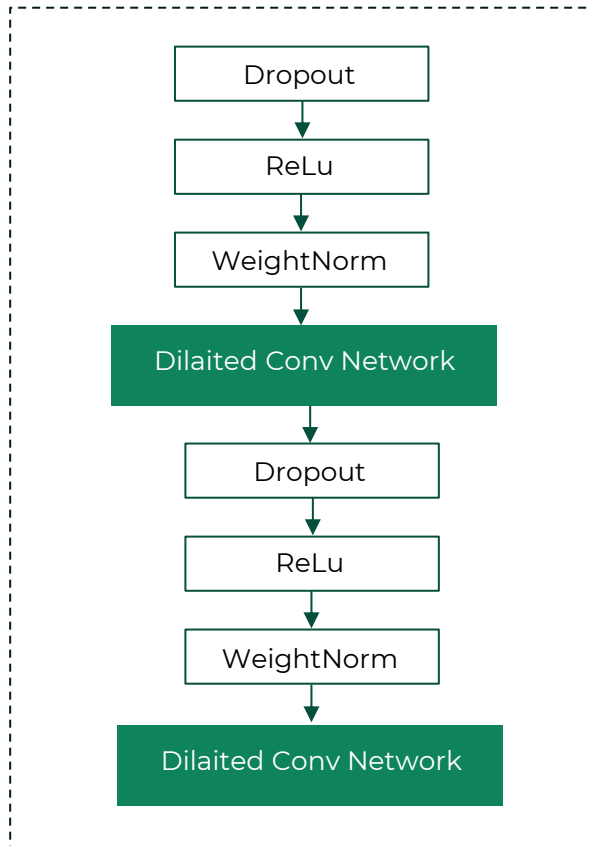


Quantum architecture
mirrors the classical
TCN

Quantum TCN



Classical TCN



Quantum architecture
mirrors the classical
TCN



**Ready for
fair benchmarking**

Benchmarking Data

Task	Input	Task
Adding Problem	Sequence of length n depth 2	Compute the sum of the two marked values
Sequential MNIST	784×1 pixel sequence	Classify the digit from the pixel sequence
JSB Chorales	88-bit piano vectors	Predict next notes in polyphonic music
Nottingham	88-bit piano vectors	Predict next notes in folk tunes

Low



High
Complexity

Experimental Setup

Task	Qubits	Learning Rate	Batch Size	Epochs	Dropout
Adding Problem	8	4×10^{-3}	32	10	0
Sequential MNIST	8	2×10^{-3}	64	20	0
JSB Chorales	8	1×10^{-3}	-	100	0.2
Nottingham	8	1×10^{-3}	-	100	0.5

Experiments were conducted using the PennyLane `default.qubit` simulator and PyTorch

Results

Task	LSTM	GRU	RNN	R-Transformer	TCN	QTCN
Adding Problem (loss)	0.164	5.3e-5	0.177	-	5.8e-5	0.139
Sequential MNIST (accuracy)	87.2	96.2	21.5	99.1	99.0	50
JSB Chorales (loss)	8.45	8.43	8.91	-	8.10	0.0026
Nottingham (loss)	3.29	3.46	4.05	2.37	3.07	3.9e-5

Results and discussion

a. Simple tasks

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¹⁹ It is interesting to note that in a similar manner, WeiNet – the only non-hybrid quantum convolutional neural network we tested – performs badly on BARS & STRIPES which we considered to be a very simple task.

« removing entanglement from a quantum model often results in as good or better performance, suggesting that “quantumness” may not be the crucial ingredient for the small learning tasks considered here »

Bowles et al. (2024). Better than Classical?

Results and discussion

b. Complex tasks

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Quantum Vs Classical TCN

1. Lower parameter count

Task	Classical TCN Parameters	QTCN Parameters
Adding Problem	~70k	~38k
Sequential MNIST	~70k	~38k
JSB Chorales	~300K	~38k
Nottingham	~1M	~38k

2. More Efficient Long-Range Dependency Capture for Complex Data

3. Quantum parallelism

However the main limitation is the dilation range constrained by the number of qubits.

Conclusion

a. Summary

- **Fair benchmarking:** Compared QTCN with a mirrored classical TCN architecture.
- **Complex data:** QTCN shows advantages on complex and structured sequential data.
- **Interpretability:** Entanglement provides a more interpretable quantum mechanism.

b. Future work

- **Model optimization:** Leverage entanglement-based interpretability to further optimize and improve QTCN..
- **Real quantum hardware:** Evaluate QTCN on real quantum devices

Thanks

Do you have any questions?

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Funded by



Deutsche
Forschungsgemeinschaft

German Research Foundation